Calc 2 – Polar Coordinates

Worksheet A

1. Convert $(2, \pi/3)$ to Cartesian coordinates.

2. Convert $(x, y) = (-1, 1)$ to polar coordinates (with $r > 0$, $0 \le \theta < 2\pi$).

3. Find the area enclosed by the circle $r = 2\sin\theta$.

4. Find the area enclosed by one petal of the rose $r = \cos(3\theta)$.

5. Find the arc length of $r = \theta$, $\theta \in [0, 1]$.

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Worksheet B

1. Convert $(\sqrt{3}, -1)$ to polar coordinates.

2. Identify the curve $r = 4\cos\theta$ in Cartesian form.

3. Find the area inside the cardioid $r = 1 + \sin\theta$.

4. Find the area inside both $r = 1$ and $r = 2\sin\theta$ (intersection region).

5. Find the slope of the curve $r = 1 + \cos\theta$ at $\theta = \pi/2$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $x = 2\cos(\pi/3) = 1$, $y = 2\sin(\pi/3) = \sqrt{3}$. $\boxed{(1, \sqrt{3})}$.

2. $r = \sqrt{2}$, $\theta = 3\pi/4$ (second quadrant). $\boxed{(\sqrt{2}, 3\pi/4)}$.

3. $r = 2\sin\theta$ is a circle of radius 1 centered at $(0,1)$. Area $= \pi(1)^2 = \boxed{\pi}$. (Verify: $A = \frac{1}{2}\int_0^\pi 4\sin^2\theta\,d\theta = 2\int_0^\pi \frac{1-\cos 2\theta}{2}\,d\theta = \pi$.)

4. One petal sweeps $\theta \in [-\pi/6, \pi/6]$. $A = \frac{1}{2}\int_{-\pi/6}^{\pi/6}\cos^2(3\theta)\,d\theta = \frac{1}{2}\cdot\frac{1}{2}\int(1+\cos 6\theta)\,d\theta = \frac{1}{4}[\theta + \frac{\sin 6\theta}{6}]_{-\pi/6}^{\pi/6} = \boxed{\pi/12}$.

5. $L = \int_0^1 \sqrt{\theta^2+1}\,d\theta = \frac{1}{2}\big[\theta\sqrt{\theta^2+1} + \sinh^{-1}\theta\big]_0^1 = \boxed{\frac{1}{2}\big(\sqrt{2} + \ln(1+\sqrt{2})\big)}$.

Worksheet B.

1. $r = \sqrt{3+1} = 2$. $\theta = \arctan(-1/\sqrt{3}) = -\pi/6$ (fourth quadrant). $\boxed{(2, -\pi/6)}$ or $(2, 11\pi/6)$.

2. Multiply by r: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow \boxed{(x-2)^2 + y^2 = 4}$. Circle of radius 2 centered at $(2,0)$.

3. $A = \frac{1}{2}\int_0^{2\pi}(1+\sin\theta)^2\,d\theta = \frac{1}{2}\int_0^{2\pi}(1 + 2\sin\theta + \sin^2\theta)\,d\theta = \frac{1}{2}(2\pi + 0 + \pi) = \boxed{3\pi/2}$.

4. Intersect: $2\sin\theta = 1 \Rightarrow \theta = \pi/6, 5\pi/6$. Region inside both: below $r = 1$ on $[\pi/6, 5\pi/6]$ and below $r = 2\sin\theta$ on $[0,\pi/6] \cup [5\pi/6, \pi]$. $A = 2\Big[\frac{1}{2}\int_0^{\pi/6}(2\sin\theta)^2\,d\theta + \frac{1}{2}\int_{\pi/6}^{\pi/2} 1\,d\theta\Big] = \int_0^{\pi/6} 4\sin^2\theta\,d\theta + \int_{\pi/6}^{\pi/2} d\theta = \big[2\theta - \sin 2\theta\big]_0^{\pi/6} + \frac{\pi}{3} = \boxed{\frac{2\pi}{3} - \frac{\sqrt{3}}{2}}$.

5. $\frac{dy}{dx} = \dfrac{f'\sin\theta + f\cos\theta}{f'\cos\theta - f\sin\theta}$ with $f = 1 + \cos\theta$, $f' = -\sin\theta$. At $\theta = \pi/2$: $f = 1$, $f' = -1$. Numerator $= -1\cdot 1 + 1\cdot 0 = -1$. Denominator $= -1\cdot 0 - 1\cdot 1 = -1$. Slope $= \boxed{1}$.